EXPERIMENT NO:06

CODE:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Create Naive Bayes model
model = GaussianNB()

# Train model
model.fit(X_train, y_train)

# Predict output
y_pred = model.predict(X_test)
```

```
# Display results
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Accuracy:", round(accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(cm)

# Graph 1: Confusion matrix
plt.figure(figsize=(6, 5))
plt.imshow(cm, cmap='Blues')
plt.title("Naive Bayes Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.xticks([0, 1, 2], iris.target_names, rotation=20)
plt.yticks([0, 1, 2], iris.target_names)
plt.colorbar()

for i in range(3):
    for j in range(3):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.tight_layout()
plt.show()
```

```
# Graph 2: Actual vs Predicted classes
plt.figure(figsize=(8, 4))
plt.plot(y_test, 'o-', label='Actual')
plt.plot(y_pred, 'x--', label='Predicted')
plt.title("Naive Bayes: Actual vs Predicted")
plt.xlabel("Test Sample")
plt.ylabel("Iris Class")
plt.yticks([0, 1, 2], iris.target_names)
plt.legend()
plt.grid(True)
plt.show()

# Graph 3: Accuracy
plt.figure(figsize=(5, 4))
plt.bar(['Naive Bayes'], [accuracy * 100])
plt.ylim(0, 105)
plt.ylabel("Accuracy (%)")
plt.title("Naive Bayes Classification Accuracy")
plt.text(0, accuracy * 100 + 2, f"{accuracy * 100:.2f}%", ha='center')
plt.show()
```

OUTPUT:

```
*** Accuracy: 96.67 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```





